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Bank-driven model for preventing double spending of digital currency transferred between multiple DLT networks using a trusted intermediary

Abstract

A method for preventing the double-spending of digital currency that transfers between multiple distributed ledger technology (DLT) networks. The method includes receiving a message including a smart contract and executing the smart contract. The method further includes detecting a transaction request to transfer the first digital currency from the first DLT network to the second DLT network. The method further includes destroying the digital currency and creating, based on the unit of fiat currency, a second digital currency of a second type on the second DLT network. The method further includes transferring, responsive to detecting the transaction request, the first digital currency from the first DLT network to the second DLT network.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS (1) This application is a continuation of, and claims priority to, U.S. patent application Ser. No. 17/870,500, “Bank-Driven Model For Preventing Double Spending Of Digital Currency Transferred Between Multiple DLT Networks Using A Trusted Intermediary,” filed Jul. 21, 2022, which is continuation of, and claims priority to, U.S. Pat. No. 11,416,848, “Bank-Driven Model for Preventing Double Spending of Digital Currency Transferred Between Multiple DLT Networks Using a Trusted Intermediary” issued Aug. 16, 2022, which is related to U.S. patent application Ser. No. 18/079,674, “Bank-Driven Model For Preventing Double Spending Of Digital Currency Coexisting On Multiple DLT Networks,” filed Dec. 12, 2022, which is continuation of, and claims priority to, U.S. Pat. No. 11,526,875, “Bank-Driven Model For Preventing Double Spending Of Digital Currency Coexisting On Multiple DLT Networks,” issued Dec. 13, 2022, which are each incorporated herein by reference in their entireties.

BACKGROUND

(1) Digital currency (also referred to as, “digital money”, “electronic money”, or “electronic currency”) is a type of currency available in digital form, as opposed to physical currency, such as banknotes and physical coins. It exhibits properties similar to physical currencies, but can allow for instantaneous transactions and borderless transfer-of-ownership. Examples of digital currency include virtual currencies, cryptocurrencies, and central bank digital currency. These currencies

may be used to buy physical goods and services, but may also be restricted to certain communities such as for use inside an online game and/or software application.

SUMMARY

(2) Aspects of the present disclosure relate generally to distributed ledger technology in the field of digital currency, and more particularly to systems and methods for preventing the double-spending of digital currency that transfers between multiple distributed ledger technology (DLT) networks.

(3) One aspect disclosed herein is directed to a method for preventing the double-spending of digital currency that transfers between multiple DLT networks. In some arrangements, the method includes receiving, by an exchange node (also referred to herein as “one or more processors”), an authorization to monitor transaction requests associated with a first DLT network. In some arrangements, the method includes detecting, by the exchange node after receiving the authorization, a transaction request to transfer a first digital currency of a first type from the first DLT network to a second DLT network, the first digital currency created on the first DLT network based on a unit of fiat currency. The method includes transferring, by the exchange node responsive to detecting the transaction request, the first digital currency from the first DLT network to the second DLT network.

(4) In another aspect, the present disclosure is directed to a system for preventing the double-spending of digital currency that transfers between multiple DLT networks. In some arrangements, the system includes one or more processors; and one or more computer-readable storage mediums storing instructions which, when executed by the one or more processors, cause the one or more processors to receive an authorization to monitor transaction requests associated with a first DLT network. In some arrangements, the system includes one or more processors and one or more computer-readable storage mediums storing instructions which, when executed by the one or more processors, cause the one or more processors to detect, after receiving the authorization, a transaction request to transfer a first digital currency of a first type from the first DLT network to a second DLT network. In some arrangements, the first digital currency created on the first DLT network based on a unit of fiat currency. In some arrangements, the system includes one or more processors and one or more computer-readable storage mediums storing instructions which, when executed by the one or more processors, cause the one or more processors to transfer, responsive to detecting the transaction request, the first digital currency from the first DLT network to the second DLT network.

(5) One aspect disclosed herein is directed to a method for preventing the double-spending of digital currency that transfers between multiple DLT networks. In some arrangements, the method includes receiving, by a node of a first DLT network of a first type, a message including a smart contract. In some arrangements, the method includes Executing, by the node of the first DLT network, the smart contract. In some arrangements, the method includes detecting, by the node of the first DLT network and via the smart contract, a transaction request to transfer a digital currency from the first DLT network to a second DLT network of a second type. In some arrangements, the method includes destroying, by the node of the first DLT network, the digital currency. In some arrangements, the method includes transferring, by the node responsive to detecting the transaction request, the digital currency from the first DLT network to the second DLT network via an exchange node that is separate from the first DLT network and the second DLT network.

(6) In another aspect, the present disclosure is directed to a non-transitory computer-readable storage medium storing instructions which, when executed by one or more processors, cause the one or more processors to perform operations including receiving an authorization to monitor transaction requests associated with a first DLT network. In some arrangements, the non-transitory computer-readable storage medium storing instructions which, when executed by one or more processors, cause the one or more processors to perform operations including detecting, after receiving the authorization, a transaction request to transfer a first digital currency of a first type from the first DLT network to a second DLT network, the first digital currency created on the first

DLT network based on a unit of fiat currency. In some arrangements, the non-transitory computer-readable storage medium storing instructions which, when executed by one or more processors, cause the one or more processors to perform operations including transferring, by the exchange node responsive to detecting the transaction request, the first digital currency from the first DLT network to the second DLT network.

(7) These and other features, together with the organization and manner of operation thereof, will become apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIG. 1 is a block diagram depicting an example environment systems and methods for preventing the double-spending of digital currency that transfers between multiple DLT networks, according to some arrangements.
- (2) FIG. 2A is a block diagram depicting an example exchange node of the environment in FIG. 1, according to some arrangements.
- (3) FIG. 2B is a block diagram depicting an example DLT node of the environment in FIG. 1, according to some arrangements.
- (4) FIG. 3 is a flow diagram depicting a method for preventing the double-spending of digital currency that transfers between multiple DLT networks from the perspective of an exchange node, according to some arrangements.
- (5) FIG. 4 is a flow diagram depicting a method for preventing the double-spending of digital currency that transfers between multiple DLT networks from the perspective of a DLT node, according to some arrangements.
- (6) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

- (7) A cryptocurrency (or crypto currency) is a digital currency designed to work as a medium of exchange that uses strong cryptography to secure financial transactions, control the creation of additional units, and verify the transfer of assets. Cryptocurrencies use decentralized control as opposed to centralized digital currency and central banking systems. The decentralized control of each cryptocurrency works through distributed ledger technology (DLT), such as a blockchain that serves as an electronic public financial transaction database.
- (8) Cryptocurrency users store their cryptocurrency in digital wallets, where the purchase, sale, and exchange transactions occur in blocks. Each block connects to the previous block by means of a code, based on blockchain technology. A cryptocurrency runs on a blockchain, which is a shared ledger or document duplicated several times across a network of computers (“nodes”). The updated document is distributed and made available to all nodes on the blockchain. Every single transaction made and the ownership of every single cryptocurrency in circulation is recorded in the blockchain. As such, the nodes of the blockchain or DLT network can prevent invalid transactions from taking place.
- (9) However, transferring digital currency such as cryptocurrency between multiple DLT networks creates problems that the digital currency (or cryptocurrency) industry has yet to solve. Firstly, a DLT network operating a first type of cryptocurrency (e.g., Bitcoin) cannot communicate with a DLT network operating a second type of digital currency (e.g., Ethereum) due to at least their incompatible communication protocols. Furthermore, even if the DLT networks could communicate with one another, there is no mechanism for preventing the double-spending of the digital currency as it traverses across the networks because each network (and its respective nodes) manages an entirely different electronic ledger. As such, the nodes of one DLT network would be

unable to verify and concretize the transactions of cryptocurrencies involving another DLT network.

(10) Accordingly, the present disclosure is directed to systems and methods for preventing the double-spending of digital currency that transfers between multiple DLT networks. In some arrangements, an exchange node converts and/or exchanges a digital currency of a first type on a first DLT network to a digital currency of a second type on a second DLT network. As further discussed herein, in some examples the exchange node may delete and/or destroy the digital currency of the first type before, or simultaneously with, re-creating the digital currency on the second DLT network as the digital currency of the second type. The exchange node may reside inside or outside of the sending/receiving DLT networks.

(11) In general, as described in the below passages and specifically in the description of FIG. 1, an issuer (e.g., issuer **140** in FIG. 1) may operate an exchange node (e.g., exchange node **130** in FIG. 1) that hosts an exchange network for participating in transactions occurring between a plurality of Distributed Ledger Technology (DLT) networks (e.g., DLT networks **102**, **104**, **106** in FIG. 1) or blockchain networks. Each DLT network is associated with a digital currency or cryptocurrency of a particular type (e.g., Bitcoin, Ether, XRP, etc.) and includes a plurality of DLT nodes (also referred to herein as, “nodes”) that are interconnected with one another to form a peer-to-peer network. The exchange network and the plurality of DLT networks are interconnected with one another via a communication network (e.g., communication network **120** in FIG. 1).

(12) To begin, the exchange node adds any number of DLT networks (e.g., the Bitcoin network, the Ethereum network, the Ripple network, etc.) that are each associated with a particular type of digital currency (e.g., a Bitcoin for the Bitcoin network, an Ether for the Ethereum network, an XRP for the Ripple network, etc.) to the exchange network in response to receiving a request from any of the DLT networks. For example, a node (e.g., any one of nodes **103a-103d** in FIG. 1) of a “first” DLT network (e.g., DLT network **102** in FIG. 1) associated with a “first” type of digital currency (e.g., a Bitcoin) may send a request to the exchange node to join the exchange network. In response to receiving the request, the exchange node adds the first DLT network to the exchange network by generating and sending a message to a node (e.g., the requesting node, a normal node, a master node) of the first DLT network to cause the node to grant permission for the exchange node to detect (e.g., monitor) the transaction requests that are sent by and/or received by a node of the first DLT network. The message also includes program code (e.g., a script, an executable) that, when executed by the node of the first DLT network, causes the node to install one or more monitoring agents (e.g., monitoring agent **150** in FIG. 1) on any or all of the nodes included in the first DLT network. The monitoring agent allows the exchange node to detect an occurrence of such transaction requests. In some arrangements, a monitoring agent that is installed on a node may be configured to intercept the transaction requests that are sent by and/or received by the node, and redirect the transaction request (or a copy of the transaction request) to the exchange node. In some instances, the message may include a smart contract, that when executed by the node, either causes the node to monitor/detect the transactions that are made by the node or allow the exchange node to monitor/detect the transactions. In some arrangements, the message may include program code that, when executed by the node of the first DLT network, causes the node to install one or more “hooks” that augment the behavior of an operating system and/or application (e.g., a digital wallet, a transaction ledger, etc.) executing on the node by intercepting function calls, messages, and/or events passed between the operating system and/or application, and redirecting the intercepted function calls, messages, and/or events to the exchange node.

(13) Continuing with the example, a node (e.g., any one of nodes **105a-105d** in FIG. 1) of a “second” DLT network (e.g., DLT network **104** in FIG. 1) associated with a “second” type of digital currency (e.g., an Ether) may also send a request to the exchange node to join the exchange network. In response to receiving the request from the second DLT network, the exchange node repeats the same process as discussed above with respect to the first DLT network, but with respect

to the second DLT network. That is, the exchange node adds the second DLT network to the exchange network by generating and sending a message to a node (e.g., the requesting node, a normal node, a master node) of the second DLT network to cause the node to grant permission for the exchange node to detect (e.g., monitor) the transaction requests that are sent by and/or received by the second DLT network. The message also includes program code (e.g., a script, an executable) to cause the node to install one or more monitoring agents (e.g., monitoring agent 152 in FIG. 1) on any or all of the nodes included in the second DLT network to allow the exchange node to detect an occurrence of such transaction requests. As discussed above, the message may include a smart contract.

(14) The exchange node then monitors the communication to and/or from any of the DLT networks that have been added to the exchange network for requests (also referred to herein as, “a transaction request”) to transfer a digital currency between nodes of the same DLT networks, as well as between nodes of different DLT networks. For example, the first DLT network may send a request (e.g., transaction request in FIG. 1) to transfer the first digital currency (e.g., a Bitcoin) from the first DLT network to the second DLT network. The exchange node detects the transaction request via the one or more monitoring agents and/or smart contracts that were previously installed and/or executed on the first DLT network (e.g., installed on one or more nodes of the first DLT network) on behalf of the exchange node, or by receiving a copy of the transaction request that was redirected by the one or more “hooks” that were previously installed on the first DLT network on behalf of the exchange node. The monitoring agents and/or smart contracts may also determine, and notify the exchange node, that the first digital currency was created based on a unit of fiat currency (e.g., collateral) held by an issuer. In response to detecting the transaction request, the exchange node destroys the first digital currency on the first DLT network to prevent a subsequent transfer of the first digital currency responsive to another transaction request (e.g., a subsequent transaction request) involving the first digital currency.

(15) After destroying the first digital currency (or concurrent with the destruction of the first digital currency), the exchange node transfers the first digital currency to the second DLT network by creating a second digital currency on the second DLT network. The second digital currency is based on the same collateral as the first digital currency. For example, the exchange node would create one or more “Ethers” on the second DLT network (e.g., an Ethereum network) based on the same unit of fiat currency that was used to create one or more “Bitcoins” on the first DLT network (e.g., a Bitcoin network). As such, the one or more Ethers and the one or more Bitcoins (prior to their destruction) would each have a value that would be equal or substantially equal to one another. The exchange node may also assign and/or attach a common serial number to each of the digital currencies to indicate that the digital currencies, despite being associated with different DLT networks, were created (or re-created) from the same collateral.

(16) As discussed above, in some instances (not all), the DLT networks that are associated with different types of digital currency may be unable to directly communicate with one another. For instance, this may be due to incompatible communication protocols. The exchange node, however, may bridge the communication between two or more DLT networks by translating their messages from one communication protocol to another communication protocol. Thus, the exchange node may function as a “trusted intermediary”. That is, the exchange node may be able to detect when a DLT network makes a transaction request to another DLT network by monitoring the exact communication it has already been assigned to translate.

(17) After transferring the first digital currency to the second DLT network, the exchange node sends a confirmation request to the second DLT network to cause the second DLT network to confirm whether the transaction has completed/approved and/or been denied. For example, a node of the second DLT network may search the second DLT network for a digital currency (e.g., the second digital currency) associated with a serial number that matches a serial number provided by the exchange node. If the second digital currency does not exist on the second DLT network, then

the second DLT sends a confirmation to the exchange node to indicate that the transaction has not been completed and/or was denied, which in turn, causes the exchange node to re-create the first digital currency on the first DLT network. However, if the second digital currency does exist on the second DLT network, then the second DLT sends a confirmation back to the first DLT network to indicate that the transaction had been completed and/or was approved. In some arrangements, instead of relying on communication from the second DLT network, the exchange node may directly search the second DTL network for the second digital currency.

(18) Thus, the exchange node is able to prevent the double spending of a digital currency on a first DLT network by destroying the digital currency on the first DLT network before re-creating the digital currency on the second DLT network.

(19) FIG. 1 is a block diagram depicting an example environment for preventing the double-spending of digital currency that transfers between multiple DLT networks, according to some arrangements. The environment **100** includes DLT networks **102**, **104**, **106** that are each associated with a digital currency or cryptocurrency of a particular type in that it hosts a public ledger that is governed by source code consisting of cryptologic and/or algorithmic protocols. DLT network **102** is associated with digital currency **110**, DLT network **104** is associated with digital currency **112**, and DLT network **106** is associated with digital currency **114**. Although FIG. 1 shows digital currency **110** as one or more “Ethers” for an Ethereum network, digital currency **112** as one or more “Bitcoins” for a Bitcoin network, and digital currency **114** as one or more “XRPs” for the Ripple network, it will be appreciated by those skilled in the art that DLT networks **102**, **104**, **106** may be any type of DLT network. As such, digital currencies **110**, **112**, **114** may be any type of digital currency that is supported by the corresponding DLT network.

(20) Each DLT network **102**, **104**, **106** includes a plurality of nodes that are interconnected with one another to form a peer-to-peer network. As shown in FIG. 1, the DLT network **102** includes nodes **103a**, **103b**, **103c**, **103d** (collectively referred to herein as, “nodes **103**”) that are interconnected with one another to form a “first” peer-to-peer network; the DLT network **104** includes nodes **105a**, **105b**, **105c**, **105d** (collectively referred to herein as, “nodes **105**”) that are interconnected with one another to form a “second” peer-to-peer network; and the DLT network **106** includes nodes **107a**, **107b**, **107c**, **107d** (collectively referred to herein as, “nodes **107**”) that are interconnected with one another to form a “third” peer-to-peer network.

(21) The environment **100** also includes an exchange node **130** that hosts an exchange network (not shown in FIG. 1) for participating in transactions occurring between any of the nodes of the DLT networks **102**, **104**, **106**. The exchange network and the DLT networks **102**, **104**, **106**, are interconnected with one another via a communication network (e.g., communication network **120** in FIG. 1). Each of the DLT networks **102**, **104**, **106**; the nodes **103**, **105**, **107**; and the exchange node **130** include hardware elements, such as one or more processors, logic devices, or circuits.

(22) As shown in FIG. 1, the exchange node is a separate entity from each of DLT networks **102**, **104**, **106**. That is, the exchange node resides and executes outside of DLT network **102**, **104**, **106**. In some arrangements, the exchange node may optionally execute on or within at least one of DLT networks **102**, **104**, **106**.

(23) The DLT networks **102**, **104**, **106** may be associated with the same or different types of digital currency. For example, the DLT network **102** may be a Bitcoin network that host a public ledger associated with one or more “Bitcoins”, the DLT network **104** may be an Ethereum network that hosts a public ledger associated with one or more “Ethers”, and the DLT network **106** may be a Ripple network that hosts a public ledger associated with one or more “XRPs”. As another example, each DLT network **102**, **104**, **106** may be a Bitcoin network that hosts a public ledger associated with one or more “Bitcoins”.

(24) A node (e.g., nodes **103a-d**, nodes **105a-d**, nodes **107a-d**) is an electronic computing device that is capable of conducting digital currency transactions (e.g., shown in FIG. 1 as “transaction requests”) with another node. The transactions that are performed by the nodes on a particular DLT

network are recorded in an electronic transaction ledger (e.g., transaction ledger **218B** in FIG. 2B). Each node may store and broadcast copies of the electronic transaction ledger to neighboring nodes to ensure that each node in the DLT network may be able to validate the transactions that occur on the DLT network via a set of consensus rules. Each node is also capable of sending a join request (shown in FIG. 1 as “join requests”) to the exchange node **130** to join the exchange network. Each node is also capable of receiving a message (e.g., shown in FIG. 1 as, “monitoring agent setup command”) to install a monitoring agent onto the node and/or the DLT network associated with the node to allow the exchange node to monitor/detect the transactions that are made by the node. For example, node **103a** may install one or more monitoring agents **150** on DLT network **102**; node **105a** may install one or more monitoring agents **152** on DLT network **104**; and node **107a** may install one or more monitoring agents **154** on DLT network **106**. Although FIG. 1 shows monitoring agent **150** executing on node **103a**, a monitoring agent **150** may be installed on any other node (e.g., node **103b**, node **103c**, node **103d**) associated with DLT network **102**. Although FIG. 1 shows monitoring agent **152** executing on node **105a**, a monitoring agent **152** may be installed on any other node (e.g., node **105b**, node **105c**, node **105d**) associated with DLT network **104**. Although FIG. 1 shows monitoring agent **154** executing on node **107a**, a monitoring agent **154** may be installed on any other node (e.g., node **107b**, node **107c**, node **107d**) associated with DLT network **106**.

(25) In some instances, the message (e.g., shown in FIG. 1 as, “monitoring agent setup command”) may include a smart contract, that when executed by the node, causes the node to monitor/detect (or allow the exchange node to monitor/detect) the transactions that are made by the node. Each node is also capable of sending a transaction confirmation (e.g., shown in FIG. 1 as, “transaction confirmation”) to the exchange node **130** indicating whether a transaction has completed and/or has been approved/denied. Each node is also capable of receiving a message (e.g., shown in FIG. 1 as, “coin creation command”) to create/add a digital currency onto the DLT network of the node. For example, the message may add an entry to the electronic transaction ledger of the node. Each node is also capable of receiving a message (e.g., shown in FIG. 1 as, “coin destruction command”) to destroy/remove a digital currency from the DLT network of the node. For example, the message may remove an entry from the electronic transaction ledger of the node. The other nodes in the DLT network become aware of the newly created digital currency or newly destroyed digital currency when the node, during the electronic ledger reconciliation process, broadcasts its electronic transaction ledger to its neighboring nodes.

(26) A node may be any number of different types of electronic computing devices (also referred to herein as, “computing device” and “electronic device”) adapted to communicate over a communication network **120**, including without limitation, a digital wallet (also known as an “e-Wallet”), a personal computer, a laptop computer, a desktop computer, a mobile computer, a tablet computer, a smart phone, an application server, a catalog server, a communications server, a computing server, a database server, a file server, a game server, a mail server, a media server, a proxy server, a virtual server, a web server, or any other type and form of computing device or combinations of devices.

(27) As used herein, a node may be a “normal” node” or as a “master node”. While a normal node and a master node are both electronic computing devices, a master node (also known as a “super node”) differs from a normal node in that a master node has more computing resources (e.g., computing power, memory resources, networking bandwidth, storage space, etc.) than a normal node. Next to validating, saving and broadcasting transactions (which are the same operations performed by a normal node), a master node may also facilitate other events on the DLT network, such as governing voting events, providing execution of protocol operations, and enforcing the laws of the corresponding DLT network. Unlike a normal node, a master node may also maintain a constant, active connection with one or more nodes of the DLT network. As such, a master node generally requires much more resources (e.g., electricity, up-time, maintenance, storage space,

memory) than a normal node.

(28) An exchange node **130** is an electronic computing device that monitors the transaction requests sent between nodes of the same DLT network and/or nodes of different DLT networks. The exchange node **130** is also capable of receiving a join request (shown in FIG. 1 as “join requests”) from a node to join an exchange network (not shown in FIG. 1) that is managed (hosted) by the exchange node **130**. The exchange node **130** is also capable of sending a message (e.g., shown in FIG. 1 as, “monitoring agent setup command”) to install a monitoring agent onto a node and/or the DLT network associated with the node. In some arrangements, the message (e.g., shown in FIG. 1 as, “monitoring agent setup command”) may include a smart contract, that when executed by the node, causes the node to monitor/detect (or allow the exchange node to monitor/detect) the transactions that are made by the node. The exchange node **130** is also capable of receiving a transaction confirmation (e.g., shown in FIG. 1 as, “transaction confirmation”) from a node indicating whether a transaction between nodes has completed. The exchange node **130** is also capable of sending a message (e.g., shown in FIG. 1 as, “coin creation command”) to a node to create/add a digital currency onto the DLT network of the node. For example, the message may create/add an entry to the electronic transaction ledger associated with the DLT network of the node. The exchange node **130** is also capable of sending a message (e.g., shown in FIG. 1 as, “coin destruction command”) to a node to destroy/remove a digital currency from the DLT network of the node. For example, the message may destroy/remove an entry from the electronic transaction ledger associated with the DLT network of the node. As discussed above, the other nodes in the DLT network update their respective electronic transaction ledgers to show the newly added digital currency or newly destroyed digital currency when the node, during the electronic ledger reconciliation process, broadcasts its electronic transaction ledger to its neighboring nodes.

(29) The environment **100** includes an issuer **140** that may develop, register, and sell securities for the purpose of financing its operations. For example, the issuer **140** may be a corporation, a bank, an investment trust, or a domestic or foreign government. The issuer **140** may make available the following types of securities: common and preferred stocks, bonds, notes, debentures, bills and derivatives. The issuer **140** maintains (e.g., stores) fiat currency **142** on behalf of an account holder of the issuer **140**. The issuer **140** includes any number of electronic computing devices (not shown in FIG. 1) for the purpose of operating/managing the exchange node **130**, and for communicating with any other electronic computing device that is connected to the communication network **120**.

(30) Fiat currency **142** (also referred to herein as, “fiat money”) money is the currency that a government has declared to be legal tender, but it is not backed by a physical commodity. The value of each country's fiat currency is determined by the supply of the currency and the demand for it to purchase goods and services. Fiat currencies **142** are backed by the credit of the economy and taxing authority of the government that issues it, as well as the faith of those who choose to use it.

(31) The environment **100** includes an exchange storage **160** for storing a plurality of associations between a plurality of serial numbers, a plurality of first digital currencies of a first type associated with a first DLT network, and a plurality of second digital currencies of a second type associated with a second DLT network. Each association of the plurality of associations corresponds to a first digital currency being in one of at least a “created” state or a “destroyed” state, and a second digital currency being in the other of the “created” state or “destroyed” state. For example, a first association may correspond with a first digital currency that was destroyed from a first DLT network, and a second digital currency that was created on a second DLT network. Each association also corresponds to a unique serial number, such that a first digital currency and a second digital currency may be associated with the same serial number. For example, FIG. 1 shows that the exchange storage **160** may store the plurality of associations by storing data (e.g., serial numbers, digital currency states, etc.) in a table. The table may include a first column (entitled, “Coin SN”) for storing a serial number associated with a digital currency, a second column

(entitled, “1st DLT Net.”) for storing state information (e.g., created, destroyed, N/A) of a first digital currency as it exists on a first DLT network, and a third column (entitled, “2nd DLT Net.”) for storing state information of the second digital currency as it exists on a second DLT network. The “created” state indicates that the digital currency was created on the corresponding DLT network. The “destroyed” state indicates that the digital currency was destroyed (or removed) from the corresponding DLT network. The “N/A” state indicates that the exchange node neither created nor destroyed the digital currency on/from the corresponding DLT network because the digital currency has not yet been transferred to the second DLT network.

(32) The communication network **120** is a local area network (LAN), a wide area network (WAN), a personal area network (PAN), or a combination of these or other networks, that interconnect the electronic computing devices (as discussed herein) and/or databases. The environment **100** may include many thousands of DLT networks **102, 104, 106**; nodes **103, 105, 107**; exchange nodes **130**; and issuers **140** that are interconnected in any arrangement to facilitate the exchange of data between such electronic computing devices.

(33) FIG. 2A is a block diagram depicting an example exchange node of the environment in FIG. 1, according to some arrangements. While various circuits, interfaces, and logic with particular functionality are shown, it should be understood that the exchange node **130** includes any number of circuits, interfaces, and logic for facilitating the functions described herein. For example, the activities of multiple circuits may be combined as a single circuit and implemented on a single processing circuit (e.g., processing circuit **202A**), as additional circuits with additional functionality are included.

(34) The exchange node **130** includes a processing circuit **202A** composed of one or more processors **203A** and a memory **204A**. A processor **203A** may be implemented as a general-purpose processor, a microprocessor, an Application Specific Integrated Circuit (ASIC), one or more Field Programmable Gate Arrays (FPGAs), a Digital Signal Processor (DSP), a group of processing components, or other suitable electronic processing components. In many arrangements, processor **203A** may be a multi-core processor or an array (e.g., one or more) of processors.

(35) The memory **204A** (e.g., Random Access Memory (RAM), Read-Only Memory (ROM), Non-volatile RAM (NVRAM), Flash Memory, hard disk storage, optical media, etc.) of processing circuit **202A** stores data and/or computer instructions/code for facilitating at least some of the various processes described herein. The memory **204A** includes tangible, non-transient volatile memory, or non-volatile memory. The memory **204A** stores programming logic (e.g., instructions/code) that, when executed by the processor **203A**, controls the operations of the exchange node **130**. In some arrangements, the processor **203A** and the memory **204A** form various processing circuits described with respect to the exchange node **130**. The instructions include code from any suitable computer programming language such as, but not limited to, C, C++, C#, Java, JavaScript, VBScript, Perl, HTML, XML, Python, TCL, and Basic. In some arrangements (referred to as “headless servers”), the exchange node **130** may omit the input/output circuit (e.g., input/output circuit **205A**), but may communicate with an electronic computing device via a network interface (e.g., network interface **206A**).

(36) The exchange node **130** includes a network interface **206A** configured to establish a communication session with a computing device for sending and receiving data over the communication network **120** to the computing device. Accordingly, the network interface **206A** includes a cellular transceiver (supporting cellular standards), a local wireless network transceiver (supporting 802.11X, ZigBee, Bluetooth, Wi-Fi, or the like), a wired network interface, a combination thereof (e.g., both a cellular transceiver and a Bluetooth transceiver), and/or the like. In some arrangements, the exchange node **130** includes a plurality of network interfaces **206A** of different types, allowing for connections to a variety of networks, such as local area networks or wide area networks including the Internet, via different sub-networks.

(37) The exchange node **130** includes an input/output circuit **205A** configured to receive user input

from and provide information to a user of the exchange node **130**. In this regard, the input/output circuit **205A** is structured to exchange data, communications, instructions, etc. with an input/output component of the exchange node **130**. Accordingly, input/output circuit **205A** may be any electronic device that conveys data to a user by generating sensory information (e.g., a visualization on a display, one or more sounds, tactile feedback, etc.) and/or converts received sensory information from a user into electronic signals (e.g., a keyboard, a mouse, a pointing device, a touch screen display, a microphone, etc.). The one or more user interfaces may be internal to the housing of the exchange node **130**, such as a built-in display, touch screen, microphone, etc., or external to the housing of the exchange node **130**, such as a monitor connected to the exchange node **130**, a speaker connected to the exchange node **130**, etc., according to various arrangements. In some arrangements, the input/output circuit **205A** includes communication circuitry for facilitating the exchange of data, values, messages, and the like between the input/output device and the components of the exchange node **130**. In some arrangements, the input/output circuit **205A** includes machine-readable media for facilitating the exchange of information between the input/output device and the components of the exchange node **130**. In still another arrangement, the input/output circuit **205A** includes any combination of hardware components (e.g., a touchscreen), communication circuitry, and machine-readable media.

(38) The exchange node **130** includes a device identification circuit **207A** (shown in FIG. 2A as device ID circuit **207A**) configured to generate and/or manage a device identifier associated with the exchange node **130**. The device identifier may include any type and form of identification used to distinguish the exchange node **130** from other computing devices. In some arrangements, a device identifier may be associated with one or more other device identifiers. In some arrangements, to preserve privacy, the device identifier may be cryptographically generated, encrypted, or otherwise obfuscated by any circuit of the exchange node **130**. In some arrangements, the exchange node **130** may include the device identifier in any communication (any of the messages in FIG. 1, e.g., a monitoring agent setup command, a coin creation command, a coin creation command, etc.) that the exchange node **130** sends to a computing device.

(39) The exchange node **130** includes a digital currency management (DCM) circuit **208A** that may be configured to receive, via the communication network **120**, a request from a node (e.g., node **103a** in FIG. 1) of a “first” DLT network (e.g., DLT network **102** in FIG. 1) to join an exchange network that is managed (e.g., hosted) by the exchange node **130**. The exchange network may be an organization of DLT networks, an association of DLT networks, or a group/collection of DLT networks; where each DLT network is associated with one another by virtue of their membership to the exchange network. In response to receiving the request, the DCM circuit **208A** may send a message (e.g., “monitoring agent setup command” in FIG. 1) to the DLT network **102**, where the message causes the DLT network **102** (e.g., one or more nodes of the DLT network **102**) to authorize the exchange node to detect (e.g., monitor) transaction requests that are associated (e.g., sent by, sent to, or received by) with the DLT network **102**. A transaction request may be a request that is sent between nodes of a DLT network (e.g., DLT network **102** in FIG. 1), or a request that is sent between a node on a first DLT network (e.g., DLT network **102** in FIG. 1) and a node of second DLT network (e.g., DLT network **104** in FIG. 1).

(40) In some arrangements, the message includes program code (e.g., a script, an executable) that, when executed by a node of a DLT network, causes the node to install one or more monitoring agents (e.g., monitoring agent **150** in FIG. 1) on any or all of the nodes included in the DLT network to allow the exchange node **130** to detect an occurrence of a transaction request. For example, node **103a** may execute the program code to install the monitoring agent **150** within its memory (e.g., memory **204B**) allowing the monitoring agent to detect (e.g., monitor) the communication to and/or from its digital currency transaction circuit (e.g., DCT circuit **210B** in FIG. 2B) and/or its network interface (e.g., network interface **206B** in FIG. 2B). As another example, node **103a** may execute the program code to install one or more monitoring agents for

monitoring systems (e.g., subsystems) of an operating system executing on the node **103a**. That is, node **103a** may install a “file system monitoring agent” configured to monitor the file system of the operating system for instructions that are sent to and/or by the file system that are indicative of a transaction request. The node **103a** may install a “network system monitoring agent” configured to monitor the network system of the operating system for instructions that are sent to and/or by the network system that are indicative of a transaction request. The node **103a** may install a “process system monitoring agent” configured to monitor the process system of the operating system for instructions that are sent to and/or by the process system that are indicative of a transaction request. The node **103a** may install a “memory management system monitoring agent” configured to monitor the memory system of the operating system for instructions that are sent to and/or by the memory system that are indicative of a transaction request.

(41) In some arrangements, the message may include program code that, when executed by a node of a DLT network, causes the node to install one or more “hooks” (not shown in FIG. 1) that augment the behavior of an operating system and/or an application executing on the node to intercept the function calls, messages, and/or events passed between software components (e.g., a digital wallet) executing on the node and/or a transaction ledger (e.g., transaction ledger **218** in FIG. 2B) stored on the node and to redirect the intercepted function calls, messages, and/or events to the exchange node **130**. The exchange node **130** may then determine if the node has sent and/or received a transaction request based on analyzing and/or processing the redirected communication (e.g., the function calls, the messages, and/or the events). The operating system and/or application executing on the node may grant permission for the exchange node **130** to insert the hook into the operating system and/or application when it sends the join request (shown in FIG. 1 as “join requests”) to the exchange node **130**.

(42) In some arrangements, the message (e.g., monitoring agent setup command in FIG. 1) may include a smart contract, that when executed by the node, causes the node (or the exchange node) of the DLT network to monitor/detect the transactions that are made by the node. That is, a smart contract is a self-executing contract where the terms and conditions are defined and enforced using software. The node may store the smart contract on the blockchain. When a transaction request is sent or received by the node on the DLT network, the smart contract may execute to notify the exchange node of the transaction request.

(43) The message may cause the node (e.g., node **103a**) to send the message (or copies thereof) to other nodes in the DLT network **102**, thereby causing those nodes to also install their own monitoring agent **150** and/or execute their own copy of the smart contract to allow the exchange node to detect (e.g., monitor) their respective digital currency transaction circuits. In some arrangements, the message causes the node (e.g., node **103a**) to send the message (or copies thereof) to only the “master nodes” (as discussed herein) that are operating on the DLT network **102**, which in turn, causes the master node to install the monitoring agent **150** within the memory (e.g., memory **204B** in FIG. 2B) of the master node and/or execute its own copy of the smart contract. The master node's installation of the monitoring agent **150** and/or execution of the smart contract allows the exchange node to detect (e.g., monitor) the communication to and/or from the master node.

(44) The DCM circuit **208A** may repeat the joining process for any number of DLT networks. For example, the DCM circuit **208A** may add a “second” DLT network (e.g., DLT network **104**) and a “third” DLT network (e.g., DLT network **106**), where the one or more nodes of the second and third DLT networks install their own monitoring agents **150** and/or execute their own copy of the smart contract to allow the exchange node **130** to detect (e.g., monitor) communication to and/or from the respective nodes.

(45) The DCM circuit **208A** may be configured to detect a transaction request to transfer a digital currency from a first DLT network to a second DLT network. For example, the node **105a** of DLT network **104** may send a transaction request to node **103a** of DLT network **102** requesting to

transfer a digital currency between the DLT networks. If a monitoring agent **150** is installed on node **103a**, then the DCM circuit **208A** may detect the transaction request via the monitoring agent **150** installed on node **103a**. If a monitoring agent **150** is installed on node **105a**, then the DCM circuit **208A** may detect the transaction request via the monitoring agent **150** installed on node **105a**. If monitoring agents **150** are each installed on node **103a** and node **105a**, then the DCM circuit **208A** may detect the transaction request via either one or both of the monitoring agents **150**. Continuing with the above example, if a smart contract is executing on node **103a**, then the DCM circuit **208A** may detect the transaction request via the smart contract executing on node **103a**. If a smart contract is executing on node **105a**, then the DCM circuit **208A** may detect the transaction request via the smart contract executing on node **105a**. If smart contracts are executing on node **103a** and node **105a**, then the DCM circuit **208A** may detect the transaction request via either of the smart contracts.

(46) In some arrangements, an installed monitoring agent and/or an executing smart contract causes a node (e.g., node **103a**) to intercept the transaction request and redirect the transaction request to the DCM circuit **208A** before the node **103a** has an opportunity to process the transaction request. In response to receiving the request, the DCM circuit **208A** may process the transaction request and then send a message (not shown in FIG. 1) to allow the node **103a** to process the transaction request that its respective monitoring agent **150** and/or smart contract intercepted. The message may include the transaction request (or details thereof) that were originally sent by the node **105a** to the node **103a**.

(47) In some arrangements, the monitoring agent **150** and/or smart contract redirects a copy of the transaction request to the DCM circuit **208A**. In this instance, the node **103a** still receives the transaction request that was sent by another node **105a**, thereby allowing the node **103a** to process the transaction request without having to wait for the DCM circuit **208A** to send a message, as discussed above. As such, the node **103a** and the DCM circuit **208A** may concurrently process the transaction requests that they each receive.

(48) In some arrangements, the DCM circuit **208A** may be configured to detect a transaction request to transfer a digital currency from a first DLT network to a second DLT network based on receiving a message from the first DLT network and/or the second DLT network. For example, a DLT network that joins the exchange network may enter into an agreement with the exchange node **130** where the terms (e.g., a set of rules) of the agreement require for the nodes of the DLT network to send their respective transaction requests, or the transaction requests that the nodes detect (via any of the techniques discussed here), to the DCM circuit **208A**.

(49) In various examples, the DCM circuit **208A** may be configured to directly or indirectly destroy the first digital currency on the first DLT network in response to detecting the transaction request. For example, the DCM circuit **208A** may send a message (e.g., shown in FIG. 1 as, “coin destruction command”) to node **103a** of DLT network **102** to instruct (and cause) the node **103a** to remove one or more entries associated with the first digital currency from the electronic transaction ledger that is maintained by the node **103a**. The other nodes in the first DLT network may discover that the first digital currency has been “destroyed” when the node **103a**, during the electronic ledger reconciliation process, broadcasts its electronic transaction ledger to its neighboring nodes. By destroying the first digital currency, the DCM circuit **208** may prevent a subsequent transfer of the first digital currency in response to an occurrence of another transaction request (e.g., a subsequent transaction request) involving the first digital currency.

(50) The DCM circuit **208A** may be configured to transfer the first digital currency to the second DLT network by creating a second digital currency on the second DLT network based on the “same” collateral that the first digital currency was previously created from. That is, by creating the second digital currency based the same collateral, the DCM circuit **208A** has effectively re-created the first digital currency on the second DLT network, but in a digital currency type that is supported by the second DLT network.

(51) The DCM circuit **208A** may be configured to prevent a coexistence of the first digital currency on the first DLT network and the second digital currency on the second DLT network by destroying the first digital currency prior to creating the second digital currency. For example, the DCM circuit **208A** may detect that node **103a** of DLT network **102** sent a transaction request to node **105a** of DLT network **104** to transfer the digital currency **110** (e.g., an Ether) from DLT network **102** to DLT network **104**. In response to detecting the transaction request, the DCM circuit **208A** may destroy the digital currency **110** on the DLT network **102**. After destroying the digital currency **110**, the DCM circuit **208A** may create the digital currency **112** on the DLT network **104**. By destroying the first digital currency prior to creating the second digital currency, the DCM circuit **208A** prevents the double spending of the first digital currency as a result of a subsequent transaction request.

(52) The DCM circuit **208A** may be configured to create a digital currency on any DLT network that has joined the exchange network. In some arrangements, the DCM circuit **208A** creates a second digital currency on a second DLT network based on the “same” collateral used to create a first digital currency on a first DLT network. For example, the exchange node **130** would create one or more “Ethers” on the DLT network **104** (e.g., a Bitcoin network) based on the same unit of fiat currency that was used to create the one or more “Bitcoins” on DLT network **102** (e.g., an Ethereum network). As such, the one or more Ethers and the one or more Bitcoins (prior to their destruction) would each have a value that would be equal or substantially equal to one another.

(53) The DCM circuit **208A** may be configured to determine the type of digital currency to create for a DLT network by determining a network type that is associated with a DLT network. For example, the DCM circuit **208A** may send a request to node **103a** of DLT network **102** to request for node **103a** to return the network type (e.g., a Bitcoin network, an Ethereum network, a Ripple network, etc.) that is associated with DLT network **102**. In response, the node **103a** may send the network type to the DCM circuit **208A**. As another example, the DCM circuit **208A** may determine the network type that is associated with DLT network **102** by traversing and/or analyzing a node (e.g., node **103a**) of the DLT network to acquire (e.g., gather, collect, etc.) characteristics of the node that indicate the type of network that the node is operating on and/or operating within.

(54) The DCM circuit **208A** may be configured to assign and/or attach a common (“global”) serial number to each of the digital currency (e.g., also referred to herein as, “digital coins” or “coins”) that it creates based on the same collateral to indicate that the digital currency were created from the same collateral. For example, the DCM circuit **208A** may create a digital currency **110** (e.g., an Ether) on DLT network **102** based on a unit of fiat currency **142** and attach a serial number to the digital currency **110**. The DCM circuit **208A** may then create a digital currency **112** (e.g., a Bitcoin) on DLT network **104** based on the unit of fiat currency **142** and attach the “same” serial number to the digital currency **112**.

(55) As another example, the DCM circuit **208A** may destroy a digital currency **110** (e.g., an Ether) that was created by the exchange node **130** (or any other computing device, such as a cryptocurrency exchange) on DLT network **102** based on a unit of fiat currency **142**. The DCM circuit **208A** may then create a digital currency **112** (e.g., a Bitcoin) on DLT network **104** based on the “same” unit of fiat currency **142** and attach a serial number to the digital currency **112**. The DCM circuit **208A** may then destroy the digital currency **112**, and re-create the digital currency **110** on the first DLT network based on the “same” unit of fiat currency **142** and attach the “same” serial number to the digital currency **110** that was previously attached to the digital currency **112**.

(56) In some arrangements, the DCM circuit **208A** may generate a serial number that is “unique” by randomly generating the serial number. In some arrangements, a serial number may be unique, in that it is associated with collateral, where no other serial numbers associated with any of the DLT networks **102**, **104**, **106** are associated with the same collateral. In some arrangements, a serial number may be unique in that it is associated with only one DLT network. In some arrangements, the DCM circuit **208A** may retrieve the serial number from a database or storage (e.g., exchange

storage **160** in FIG. 1).

(57) The DCM circuit **208A** may be configured to store the serial numbers that it assigns and/or attaches to each of the digital currency in a storage (e.g., exchange storage **160** in FIG. 1). The DCM circuit **208** maintains, in the storage, a plurality of associations between a plurality of serial numbers, a plurality of first digital currencies of a first type associated with a first DLT network, and a plurality of second digital currencies of a second type associated with a second DLT network. Each association of the plurality of associations corresponds to a first digital currency being in a “created” state or a “destroyed” state and a second digital currency being in the other of the “created” state or “destroyed” state. For example, a first association may correspond with a first digital currency that was destroyed from a first DLT network and a second digital currency that was created on a second DLT network. Each association also corresponds to a unique serial number, such that a plurality of digital currencies (e.g., a first digital currency, a second digital currency) may be associated with the same serial number, regardless if the digital currencies may be associated with different DLT networks.

(58) The DCM circuit **208A** may be configured to send a confirmation request to the second DLT network, to cause the second DLT network to generate a confirmation indicating whether the transaction has completed and/or been approved/denied. The DCM circuit **208A** may be configured to receive the confirmation from the second DLT network. In some arrangements, the confirmation request includes a serial number (e.g., a common serial number, a global serial number) that is associated with the first digital currency. In some arrangements, the confirmation request causes the second DLT network to generate the confirmation by searching the second DLT network for a digital currency that is associated (e.g., assigned, attached) with the serial number. For example, the DCM circuit **208A** may send a confirmation request to the DLT network **104** where the confirmation request includes a first serial number (e.g., 1007) that is attached to the digital currency **110**. In response to receiving the confirmation request, the DLT network **104** may search the electronic transaction ledger (e.g., transaction ledger **218B** in FIG. 2B) of one or more nodes of the DLT network **104** for a digital currency associated with the first serial number (e.g., 1007), and determine that the digital currency **112** is associated with a second serial (e.g., **1007**) that matches the first serial number.

(59) The DCM circuit **208A** may be configured to confirm whether a transaction has completed and/or been approved/denied by determining a presence or an absence of the second digital currency on the second DLT network. For example, the DCM circuit **208A** may search the electronic transaction ledger (e.g., transaction ledger **218B** in FIG. 2B) of one or more nodes of the DLT network **104** for a digital currency associated with the first serial number and determine that the digital currency **112** is associated with a second serial number (e.g., 1007) that matches the “first serial number. In some arrangements, the DCM circuit **208A** searches the electronic transaction ledger of the one or more nodes **105** of the DLT network **104** via the one or more monitoring agents **150** that were installed on the one or more nodes **105** of the DLT network **104**.

(60) In response to determining that the transaction has completed and/or been approved, the DCM circuit **208A** may send a message to the first DLT network and/or the second DLT network indicating an approval of the transaction request.

(61) In response to determining that the transaction has not completed and/or has been denied, the DCM circuit **208A** may re-create the first digital currency on the first DLT network based on the same unit of fiat currency. That is, first digital currency may be re-created based on the unit of fiat currency that was initially used to create the first digital currency. In some arrangements, the DCM circuit **208A** causes the node to broadcast the electronic transaction ledger (now updated) to other nodes in the DLT network **102**.

(62) The exchange node **130** includes a bus (not shown), such as an address/data bus or other communication mechanism for communicating information, which interconnects circuits and/or subsystems of the exchange node **130**. In some arrangements, the exchanges node **130** may include

one or more of any such circuits and/or subsystems.

(63) In some arrangements, some or all of the circuits of the exchange node **130** may be implemented with the processing circuit **202A**. For example, the DCM circuit **208A** may be implemented as a software application stored within the memory **204A** and executed by the processor **203A**. Accordingly, such arrangement can be implemented with minimal or no additional hardware costs. In some arrangements, any of these above-recited circuits rely on dedicated hardware specifically configured for performing operations of the circuit.

(64) FIG. **2B** is a block diagram depicting an example node of the environment in FIG. **1**, according to some arrangements. That is, any of the nodes (e.g., nodes **103**, **105**, **107**) in FIG. **1** may be a DLT node **201** in FIG. **2B**. While various circuits, interfaces, and logic with particular functionality are shown, it should be understood that DLT node **201** includes any number of circuits, interfaces, and logic for facilitating the functions described herein. For example, the activities of multiple circuits may be combined as a single circuit and implemented on a single processing circuit (e.g., processing circuit **202B**), as additional circuits with additional functionality are included.

(65) The DLT node **201** includes a processing circuit **202B** composed of one or more processors **203A** and a memory **204B**. The processing circuit **202B** includes identical or nearly identical functionality as processing circuit **202A** in FIG. **2A**, but with respect to circuits and/or subsystems of the DLT node **201** instead of circuits and/or subsystems of the exchange node **130**.

(66) The memory **204B** (e.g., Random Access Memory (RAM), Read-Only Memory (ROM), Non-volatile RAM (NVRAM), Flash Memory, hard disk storage, optical media, etc.) of processing circuit **202B** stores data and/or computer instructions/code for facilitating at least some of the various processes described herein. The memory **204B** includes identical or nearly identical functionality as memory **204A** in FIG. **2A**, but with respect to circuits and/or subsystems of the DLT node **201** instead of circuits and/or subsystems of the exchange node **130**.

(67) The DLT node **201** includes a network interface **206B** configured to establish a communication session with a computing device for sending and receiving data over the communication network **120** to the computing device. Accordingly, the network interface **206B** includes identical or nearly identical functionality as network interface **206A** in FIG. **2A**, but with respect to circuits and/or subsystems of DLT node **201** instead of circuits and/or subsystems of the exchange node **130**.

(68) The DLT node **201** includes an input/output circuit **205B** configured to receive user input from and provide information to a user. In this regard, the input/output circuit **205B** is structured to exchange data, communications, instructions, etc. with an input/output component of the DLT node **201**. The input/output circuit **205B** includes identical or nearly identical functionality as input/output circuit **205A** in FIG. **2A**, but with respect to circuits and/or subsystems of the DLT node **201** instead of circuits and/or subsystems of the exchange node **130**.

(69) The DLT node **201** includes a device identification circuit **207B** (shown in FIG. **2B** as device ID circuit **207B**) configured to generate and/or manage a device identifier associated with the DLT node **201**. The device ID circuit **207B** includes identical or nearly identical functionality as device ID circuit **207A** in FIG. **2A**, but with respect to circuits and/or subsystems of the DLT node **201** instead of circuits and/or subsystems of the exchange node **130**.

(70) The DLT node **201** includes a digital currency transaction (DCT) circuit **210B** composed of digital currency source code **215B** and an electronic transaction ledger (shown in FIG. **1** as, “transaction ledger **218B**”). The digital currency source code **215B** may be stored in memory **204B**, which may be accessed by and/or run on processor **203B**. The transaction ledger **218B** (shown in FIG. **1** as, “transaction ledger **218B**”) may be stored on the same and/or different processor readable memory, which may be accessible by processor **203A** when running the digital currency source code **215B**. In some arrangements, the transaction ledger **218B** on a first node (e.g., node **103a** in FIG. **1**) of a DLT network corresponds with the transaction ledger of one or more nodes

within the DLT network, to the extent that the nodes have synchronized/updated their electronic transaction ledgers (e.g., received the latest transactions via a download or during a reconciliation process). Accordingly, the electronic transaction ledger **115** may be a public ledger.

(71) A DLT node **201** of a “first” DLT network (e.g., DLT network **102** in FIG. **1**) may be configured to send, via the communication network **120**, a request to an exchange node (e.g., exchange node **130** in FIG. **1**) to join an exchange network that is managed (hosted) by the exchange node **130**.

(72) The DLT node **201** may be configured to receive a message from the exchange node **130** that causes the DLT node **201** to authorize the exchange node **130** to detect (e.g., monitor) transaction requests that are associated (e.g., sent by, sent to, or received by) with the DLT network **102**. In some arrangements, the message includes program code (e.g., a script, an executable) that, when executed by a node of a DLT network, causes the node to install one or more monitoring agents (e.g., monitoring agent **150** in FIG. **1**) on any or all of the nodes included in the DLT network to allow the exchange node **130** to detect an occurrence of a transaction request.

(73) In some arrangements, the message may include a smart contract, that when executed by the DLT node **201**, causes the DLT node **201** of the DLT network to monitor/detect the transactions that are made by the DLT node **201**. The DLT node **201** may store the smart contract on the blockchain. When a transaction request is sent or received by the DLT node **201** on the DLT network, the smart contract may execute to notify the exchange node of the transaction request.

(74) In some arrangements, the message may cause the DLT node **201** to send the message (or copies thereof) to other nodes in the DLT network **102**, thereby causing those nodes to also install their own monitoring agent **150** to allow the exchange node to detect (e.g., monitor) their respective digital currency transaction circuits. In some arrangements, the message causes the DLT node **201** to send the message (or copies thereof) to only the “master nodes” (as discussed herein) that are operating on the DLT network **102**, which in turn, causes the master node to install the monitoring agent **150** within the memory (e.g., memory **204B** in FIG. **2B**) of the master node. The monitoring agent **150** that is installed on the master node allows the exchange node to detect (e.g., monitor) the communication to and/or from the master node.

(75) The DLT node **201** may be configured to receive a message (e.g., coin creation command in FIG. **1**) causing the DLT node **201** to create a digital currency based on a unit of fiat currency (e.g., fiat currency **142** in FIG. **1**) that is identified in the message.

(76) The DLT node **201** may be configured to receive a request from the exchange node **130** to return the network type (e.g., a Bitcoin network, an Ethereum network, a Ripple network, etc.) that is associated with DLT network of the DLT node **201**. In response, the DLT node **201** sends the network type to the exchange node **130**.

(77) As discussed herein, the DLT node **201** may be configured to install one or more monitoring agents (e.g., monitoring agent **150** in FIG. **1**) within its memory to allow the monitoring agent to detect an occurrence of a transaction request associated (e.g., sent to, sent by, received by) with the DLT node **201**. The DLT node **201** may be configured to execute a smart contract that causes the DLT node **201** to monitor/detect (or allow the exchange node to monitor/detect) the transactions that are made by the DLT node **201**.

(78) In some arrangements, the DLT node **201** may be configured to detect a transaction request by determining that a copy of an electronic transaction ledger that was broadcasted by a neighboring node (e.g., node **103b** in FIG. **1**) includes one or more transaction entries that are different than the transaction entries of an electronic transaction ledger (e.g., transaction ledger **218B** in FIG. **2B**) previously stored on the node. For example, node **103a** of DLT network **102** may receive a transaction request from node **105a** of DLT network **104** to transfer a digital currency **110** (e.g., an Ether) from DLT network **102** to DLT network **104**. In response to receiving the transaction request, the DLT network **102** may post (e.g., write, record) the transaction request as an entry on its electronic transaction ledger (e.g., transaction ledger **218B** on node **103a**) and broadcast a copy

of its electronic transaction ledger (now updated) to node **103c** in DLT network for validation. In response to receiving the copy of the electronic transaction ledger, the node **103c** may compare the entries of the received electronic transaction ledger with the transaction entries of its own electronic transaction ledger (e.g., transaction ledger **218B** on node **103c**). The node **103c** determines that a transaction request occurred if it determines, as a result of the comparison, that the electronic transaction ledgers are different. The node then sends a message to the exchange node **130** to indicate that the node detected a transaction request, where the message includes the details (e.g., transacting parties, transacting amount, etc.) associated with the transaction request.

(79) The DLT node **201** may be configured to receive a confirmation request from the exchange node **130** to return a confirmation indicating whether the transfer of the digital currency from the first DLT network to the second DLT network has completed (and/or has been approved/denied). In response to receiving the request, the DLT node **201** determines if the transfer has completed (and/or approved/denied) and sends a confirmation to the exchange node **130** indicating this determination. In some arrangements, the confirmation request causes the DLT node **201** to generate the confirmation by searching the DLT network of the DLT node **201** for a digital currency that is associated (e.g., assigned, attached) with a “common” or “global” serial number. In some arrangements, the DLT node **201** may search the electronic transaction ledger (e.g., transaction ledger **218B** in FIG. 2B) of one or more nodes of the DLT network for a digital currency associated with the serial number.

(80) The DLT node **201** includes a bus (not shown), such as an address/data bus or other communication mechanism for communicating information, which interconnects circuits and/or subsystems (e.g., digital currency transaction circuit **210B**) of the DLT node **201**. In some arrangements, the DLT node **201** may include one or more of any such circuits and/or subsystems.

(81) In some arrangements, some or all of the circuits of the DLT node **201** may be implemented with the processing circuit **202B**. For example, any of the DLT node **201** may be implemented as a software application stored within the memory **204B** and executed by the processor **203B**. Accordingly, such arrangement can be implemented with minimal or no additional hardware costs. In some arrangements, any of these above-recited circuits rely on dedicated hardware specifically configured for performing operations of the circuit.

(82) FIG. 3 is a flow diagram depicting a method for preventing the double-spending of digital currency that transfers between multiple DLT networks from the perspective of an exchange node, according to some arrangements. Additional, fewer, or different operations may be performed in the method depending on the particular arrangement. In some arrangements, some or all operations of method **300** may be performed by one or more processors executing on one or more computing devices, systems, or servers. In some arrangements, method **300** may be performed by one or more exchange nodes, such as exchange node **130** in FIG. 1. Each operation may be re-ordered, added, removed, or repeated.

(83) As shown in FIG. 3, the method **300** includes the operation **302** of receiving, by one or more processors, an authorization to monitor transaction requests associated with a first distributed ledger technology (DLT) network. The method also includes the operation **304** of detecting, by the one or more processors after receiving the authorization, a transaction request to transfer a first digital currency of a first type from the first DLT network to a second DLT network. In some arrangements, the first digital currency created on the first DLT network based on a unit of fiat currency. The method also includes the operation **306** of transferring, by the one or more processors responsive to detecting the transaction request, the first digital currency from the first DLT network to the second DLT network.

(84) FIG. 4 is a flow diagram depicting a method for preventing the double-spending of digital currency that transfers between multiple DLT networks from the perspective of a DLT node, according to some arrangements. Additional, fewer, or different operations may be performed in the method depending on the particular arrangement. In some arrangements, some or all operations of

method **400** may be performed by one or more processors executing on one or more computing devices, systems, or servers. In some arrangements, some or all operations of method **400** may be performed by one or more nodes, such as any of nodes **103**, **105**, **107** in FIG. 1. Each operation may be re-ordered, added, removed, or repeated.

(85) As shown in FIG. 4, the method **400** includes the operation **402** of receiving, by a node of a first DLT network of a first type, a message including a smart contract. In some arrangements, the method includes the operation **404** of executing, by the node of the first DLT network, the smart contract. In some arrangements, the method includes the operation **406** of detecting, by the node of the first DLT network and via the smart contract, a transaction request to transfer a digital currency from the first DLT network to a second DLT network of a second type. In some arrangements, the method includes the operation **408** of destroying, by the node of the first DLT network, the digital currency. In some arrangements, the method includes the operation **410** of transferring, by the node of the first DLT network, the digital currency from the first DLT network to the second DLT network via an exchange node. In some arrangements, the exchange node is separate from the first DLT network and the second DLT network.

(86) The arrangements described herein have been described with reference to drawings. The drawings illustrate certain details of specific arrangements that implement the systems, methods and programs described herein. However, describing the arrangements with drawings should not be construed as imposing on the disclosure any limitations that may be present in the drawings.

(87) It should be understood that no claim element herein is to be construed under the provisions of 35 U.S.C. § 112(f), unless the element is expressly recited using the phrase “means for.”

(88) As used herein, the term “circuit” may include hardware structured to execute the functions described herein. In some arrangements, each respective “circuit” may include machine-readable media for configuring the hardware to execute the functions described herein. The circuit may be embodied as one or more circuitry components including, but not limited to, processing circuitry, network interfaces, peripheral devices, input devices, output devices, sensors, etc. In some arrangements, a circuit may take the form of one or more analog circuits, electronic circuits (e.g., integrated circuits (IC), discrete circuits, system on a chip (SOCs) circuits, etc.), telecommunication circuits, hybrid circuits, and any other type of “circuit.” In this regard, the “circuit” may include any type of component for accomplishing or facilitating achievement of the operations described herein. For example, a circuit as described herein may include one or more transistors, logic gates (e.g., NAND, AND, NOR, OR, XOR, NOT, XNOR, etc.), resistors, multiplexers, registers, capacitors, inductors, diodes, wiring, and so on).

(89) The “circuit” may also include one or more processors communicatively coupled to one or more memory or memory devices. In this regard, the one or more processors may execute instructions stored in the memory or may execute instructions otherwise accessible to the one or more processors. In some arrangements, the one or more processors may be embodied in various ways. The one or more processors may be constructed in a manner sufficient to perform at least the operations described herein. In some arrangements, the one or more processors may be shared by multiple circuits (e.g., circuit A and circuit B may comprise or otherwise share the same processor which, in some example arrangements, may execute instructions stored, or otherwise accessed, via different areas of memory). Alternatively or additionally, the one or more processors may be structured to perform or otherwise execute certain operations independent of one or more co-processors. In other example arrangements, two or more processors may be coupled via a bus to enable independent, parallel, pipelined, or multi-threaded instruction execution. Each processor may be implemented as one or more general-purpose processors, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), digital signal processors (DSPs), or other suitable electronic data processing components structured to execute instructions provided by memory. The one or more processors may take the form of a single core processor, multi-core processor (e.g., a dual core processor, triple core processor, quad core processor, etc.),

microprocessor, etc. In some arrangements, the one or more processors may be external to the apparatus, for example the one or more processors may be a remote processor (e.g., a cloud based processor). Alternatively or additionally, the one or more processors may be internal and/or local to the apparatus. In this regard, a given circuit or components thereof may be disposed locally (e.g., as part of a local server, a local computing system, etc.) or remotely (e.g., as part of a remote server such as a cloud based server). To that end, a “circuit” as described herein may include components that are distributed across one or more locations.

(90) An exemplary system for implementing the overall system or portions of the arrangements might include a general purpose computing computers in the form of computers, including a processing unit, a system memory, and a system bus that couples various system components including the system memory to the processing unit. Each memory device may include non-transient volatile storage media, non-volatile storage media, non-transitory storage media (e.g., one or more volatile and/or non-volatile memories), etc. In some arrangements, the non-volatile media may take the form of ROM, flash memory (e.g., flash memory such as NAND, 3D NAND, NOR, 3D NOR, etc.), EEPROM, MRAM, magnetic storage, hard discs, optical discs, etc. In other arrangements, the volatile storage media may take the form of RAM, TRAM, ZRAM, etc.

(91) Combinations of the above are also included within the scope of machine-readable media. In this regard, machine-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions. Each respective memory device may be operable to maintain or otherwise store information relating to the operations performed by one or more associated circuits, including processor instructions and related data (e.g., database components, object code components, script components, etc.), in accordance with the example arrangements described herein.

(92) It should also be noted that the term “input devices,” as described herein, may include any type of input device including, but not limited to, a keyboard, a keypad, a mouse, joystick or other input devices performing a similar function. Comparatively, the term “output device,” as described herein, may include any type of output device including, but not limited to, a computer monitor, printer, facsimile machine, or other output devices performing a similar function.

(93) Any foregoing references to currency or funds are intended to include fiat currencies, non-fiat currencies (e.g., precious metals), and math-based currencies (often referred to as cryptocurrencies). Examples of math-based currencies include Bitcoin, Ethereum, Litecoin, Dogecoin, and the like.

(94) It should be noted that although the diagrams herein may show a specific order and composition of method steps, it is understood that the order of these steps may differ from what is depicted. For example, two or more steps may be performed concurrently or with partial concurrence. Also, some method steps that are performed as discrete steps may be combined, steps being performed as a combined step may be separated into discrete steps, the sequence of certain processes may be reversed or otherwise varied, and the nature or number of discrete processes may be altered or varied. The order or sequence of any element or apparatus may be varied or substituted according to alternative arrangements. Accordingly, all such modifications are intended to be included within the scope of the present disclosure as defined in the appended claims. Such variations will depend on the machine-readable media and hardware systems chosen and on designer choice. It is understood that all such variations are within the scope of the disclosure. Likewise, software and web implementations of the present disclosure could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various database searching steps, correlation steps, comparison steps and decision steps.

(95) It is also understood that any reference to an element herein using a designation such as “first,” “second,” and so forth does not generally limit the quantity or order of those elements. Rather, these designations can be used herein as, a convenient means of distinguishing between two or

more elements or instances of an element. Thus, a reference to first and second elements does not mean that only two elements can be employed, or that the first element must precede the second element in some manner.

(96) The foregoing description of arrangements has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form disclosed, and modifications and variations are possible in light of the above teachings or may be acquired from this disclosure. The arrangements were chosen and described in order to explain the principals of the disclosure and its practical application to enable one skilled in the art to utilize the various arrangements and with various modifications as are suited to the particular use contemplated. Other substitutions, modifications, changes and omissions may be made in the design, operating conditions and arrangement of the arrangements without departing from the scope of the present disclosure as expressed in the appended claims.

Claims

1. A method, comprising: detecting, by a node of a first distributed ledger technology (DLT) network and via a smart contract, a transaction request to transfer a first digital currency from the first DLT network of a first type to a second DLT network of a second type, wherein the transaction request is received by the node of the first DLT from the first DLT network or the second DLT network; receiving, by the node of the first DLT network, a message from an exchange node indicating that the first digital currency is to be destroy; transferring, by the node of the first DLT network, the first digital currency from the first DLT network to the second DLT network via the exchange node, wherein the transferring comprises: in response to receiving the message, destroying, by the node of the first DLT network, the first digital currency from the first DLT network; creating, by the node of the first DLT network, a second digital currency of the second type on the second DLT network.
2. The method of claim 1, further comprising sending, by the node of the first DLT network via the exchange node and responsive to transferring the first digital currency from the first DLT network to the second DLT network, a confirmation request to the second DLT network.
3. The method of claim 1, further comprising receiving, by the node of the first DLT network via the exchange node, the confirmation from the second DLT network indicating whether the transfer of the first digital currency from the first DLT network to the second DLT network has completed.
4. The method of claim 1, further comprising receiving, by the node from the first DLT network from the exchange node, a destruction command to destroy the first digital currency from the first DLT network.
5. The method of claim 1, wherein creating the second digital currency of the second type on the second DLT network comprises attaching, by the node of the first DLT network via the exchange node, a serial number to the second digital currency.
6. The method of claim 1, wherein the first digital currency is associated with a created state or destroyed state and the second digital currency is associated with the other of the created state or the destroyed state.
7. The method of claim 1, wherein the first digital currency of the first type and the second digital currency of the second type are different types of digital currency.
8. A system of a first distributed ledger technology (DLT) network, comprising: at least one processing circuit configured to: detect, via a smart contract, a transaction request to transfer a first digital currency from the first DLT network of a first type to a second DLT network of a second type, wherein the transaction request is received by the system of the first DLT from the first DLT network or the second DLT network; receive a message from an exchange node indicating that the first digital currency is to be destroy; transfer the first digital currency from the first DLT network to the second DLT network via the exchange node, wherein the transferring comprises: in response

to receiving the message, destroy the first digital currency from the first DLT network; create a second digital currency of the second type on the second DLT network.

9. The system of claim 8, wherein the at least one circuit is further configured to send via the exchange node and responsive to transferring the first digital currency from the first DLT network to the second DLT network, a confirmation request to the second DLT network.
 10. The system of claim 8, wherein the at least one circuit is further configured to receiving, by the node of the first DLT network via the exchange node, the confirmation from the second DLT network indicating whether the transfer of the first digital currency from the first DLT network to the second DLT network has completed.
 11. The system of claim 8, wherein the at least one circuit is further configured to receive from the exchange node a destruction command to destroy the first digital currency from the first DLT network.
 12. The system of claim 8, wherein creating the second digital currency of the second type on the second DLT network comprises attaching, by the node of the first DLT network via the exchange node, a serial number to the second digital currency.
 13. The system of claim 8, wherein the first digital currency is associated with a created state or destroyed state and the second digital currency is associated with the other of the created state or the destroyed state.
 14. The system of claim 8, wherein the first digital currency of the first type and the second digital currency of the second type are different types of digital currency.
 15. One or more non-transitory computer-readable storage media having instructions stored thereon that, when executed by one or more processing circuits, cause the plurality of processing circuits to: detect, via a smart contract, a transaction request to transfer a first digital currency from the first DLT network of a first type to a second DLT network of a second type, wherein the transaction request is received by the system of the first DLT from the first DLT network or the second DLT network; receive a message from an exchange node indicating that the first digital currency is to be destroy; transfer the first digital currency from the first DLT network to the second DLT network via the exchange node, wherein the transferring comprises: in response to receiving the message, destroy the first digital currency from the first DLT network; create a second digital currency of the second type on the second DLT network.
 16. The non-transitory computer-readable storage media of claim 15, wherein the one or more processing circuits are further configured to send via the exchange node and responsive to transferring the first digital currency from the first DLT network to the second DLT network, a confirmation request to the second DLT network.
 17. The non-transitory computer-readable storage media of claim 15, wherein the one or more processing circuits are further configured to receiving, by the node of the first DLT network via the exchange node, the confirmation from the second DLT network indicating whether the transfer of the first digital currency from the first DLT network to the second DLT network has completed.
 18. The non-transitory computer-readable storage media of claim 15, wherein the one or more processing circuits are further configured to receive from the exchange node a destruction command to destroy the first digital currency from the first DLT network.
 19. The non-transitory computer-readable storage media of claim 15, wherein creating the second digital currency of the second type on the second DLT network comprises attaching, by the node of the first DLT network via the exchange node, a serial number to the second digital currency.
 20. The non-transitory computer-readable storage media of claim 15, wherein the first digital currency is associated with a created state or destroyed state and the second digital currency is associated with the other of the created state or the destroyed state.
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